

MTU_Value*Service* **Technical Documentation**

Engine Series 183

Base: Mercedes-Benz OM 400

6 V 183 ...1 OM 421 - 424 LA

8 V 183... 1 OM 421 - 424 LA

10 V 183... 1 OM 421 - 424 LA

12 V 183... 1 OM 421 - 424 LA

Operating Instructions

M015130/10E

-Euro 0-



Printed in Germany

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Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

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Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

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Para evitar falhas ou danos durante a operação, os dizeres do manual devem ser respeitados. Quem explora o equipamento economicamente consequentemente deve colocá-lo à disposição do respectivo pessoal da conservação, e à disposição dos operadores.

Salvo alterações.

General Information

This Operation Manual is a supplement to the **Operation Manual MO 15110**



During operation of the propulsion plant, certain regulations regarding operational reliability and environmental protection must be observed.

This publication contains additional instructions regarding **operation, maintenance and troubleshooting** as well as **important information on accident prevention and environmental protection**.

The operator is also obliged to familiarize himself with the information given below on the applicable regulations.

Your engine is equipped in accordance with how it is to be installed, i.e. not all components described in this Operation Manual are fitted in your engine.

We have made every effort to emphasize the differences clearly so that you can easily find the operating and maintenance information relevant to your engine.

MTU accepts no responsibility for incorrect or illegal operation and maintenance of the propulsion plant.

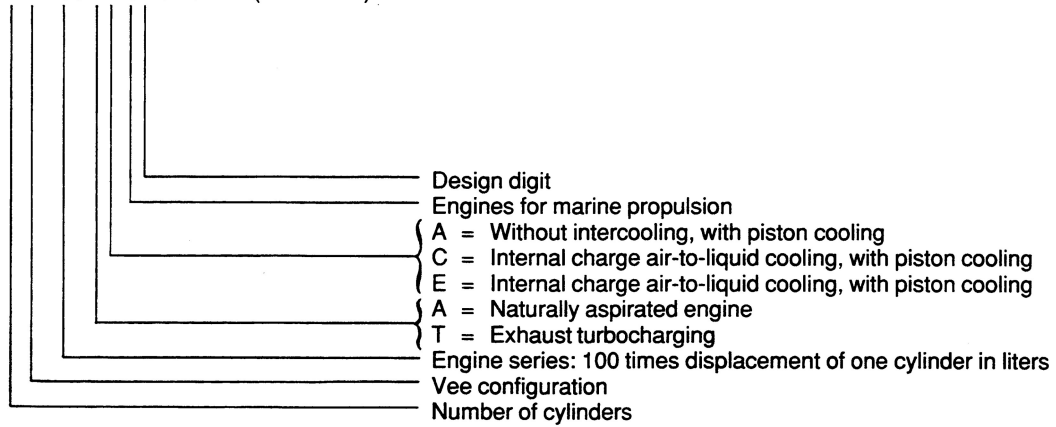


Before operating or carrying out any maintenance or repair work on this product, read all safety measures and warning notices thoroughly.

Basic safety measures and symbols are listed in the "Maintenance" chapter of this Operation Manual. Special safety instructions are included in the description of potentially hazardous tasks. Information and warning signs are also attached to the product itself.

Key to MTU Engine Designations

6V 183 AA 51/61/91 (OM421)
8V 183 AA 51/61/91 (OM422)
10V 183 AA 51/61/91 (OM423)
12V 183 AA 51/61/91 (OM424)
12V 183 TA 51/61/91 (OM424A)
12V 183 TC 51/61/91 (OM424LA)
12V 183 TE 51/61/91 (OM424LA)



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* See Operation Manual MO 15110

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* See Operation Manual MO 15110

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The components of the basic engine are described in Operation Manual MO 15110.

The engine parts and engine sub-assemblies needed for the marine engines are described and illustrated below.

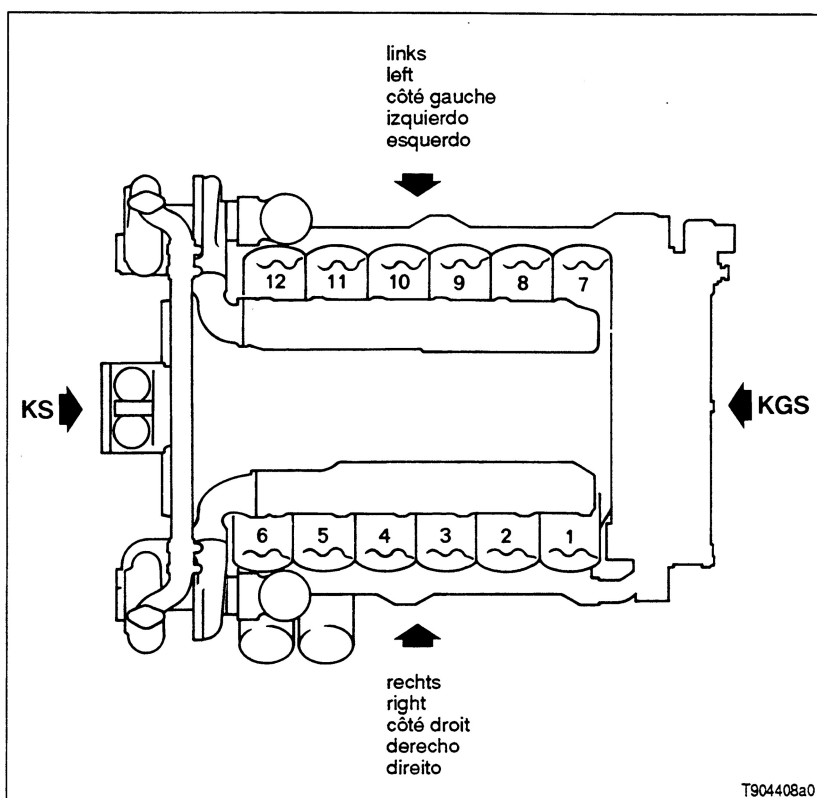
Designation of Engine Ends

according to motor vehicle standard DIN 73021

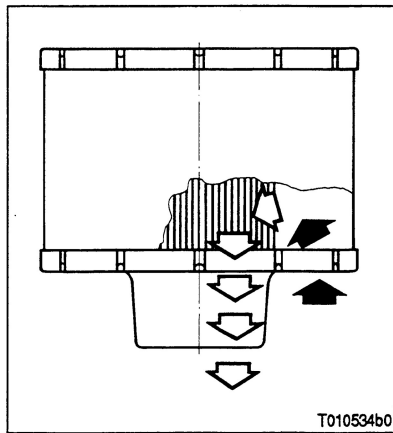
The engine is viewed from the main PTO end for the purposes of engine-end designation:

KS = main PTO end = rear

KGS = auxiliary PTO end = front



Description of Engines



Air Filter (Donaldson)

The dry-type air filter is mounted directly on the engine air intake to filter the combustion air.

The filter is made entirely of plastic and forms a unit with the integrated paper filter element.

Dust particles suspended in the air drawn through the filter are retained by the paper element.

A vacuum-type restriction indicator screwed into the filter housing allows the filter's degree of contamination to be monitored.

Maintenance:

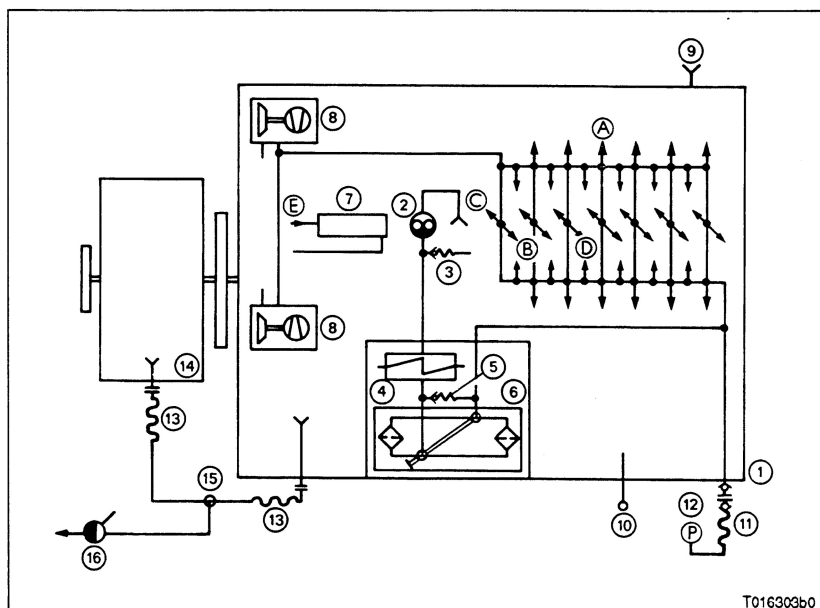
Check vacuum-type restriction indicator, see Operation Manual MO 15110.

Remove the air filter and install a replacement when the restriction indicator shows red.

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Engine Oil System

- 1 Engine
- 2 Engine oil pump(s)
- 3 Pressure relief valve (6.5 bar)
- 4 Engine oil heat exchanger
- 5 Filter bypass valve (2.7 bar)
- 6 Engine oil filters (selectable)
- 7 Fuel injection pump with engine governor
- 8 Exhaust turbocharger (12V 183 TA/TC/TE only)
- 9 Oil filler neck
- 10 Oil dipstick
- 11 Hose with quick-close connector
- 12 Oil pressure transmitter
- 13 Hose
- 14 Gearbox
- 15 Diverter valve
- 16 Handpump for oil change
- A to rockers
- B to main bearings
- C to camshaft
- D to piston cooling
- E from camshaft



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Engine Oil System

The engine oil system is a wet-sump, forced-feed system.

The 6-, 8- and 10-cylinder engines receive oil from an engine oil pump bolted to the underside of the crankcase. The oil pump is gear-driven off the crankshaft gear. A pressure relief valve with an opening pressure of 6.5 bar protects the pump.

Two oil pumps working in parallel circulate the oil in the 12-cylinder engines. The two oil pumps are bolted to the underside of the crankcase at the wall at main bearing 5 and main bearing 7. The rear oil pump has a drive gear driven by the crankshaft gear and drives the second pump via a shaft with two splined bushes and an intermediate gearbox. Two pressure relief valves with opening pressures of 6.5 bar protect the oil system against overload.

The oil pumps draw oil from the oil pan via strainers and force it through the engine oil heat exchanger and the selectable engine oil filters. Cooled and filtered, the oil then flows to the

main oil gallery. A pressure relief valve with an opening pressure of 2.7 bar opens a bypass when a filter is clogged.

From the main oil gallery, the oil flows through oilways to the crankshaft, the connecting rods, the camshaft, the injection nozzles for cam lubrication and piston cooling and to the rocker bearings. Oil leaving the rocker bearings lubricates the valves and returns to the oil pan via oilways in the cylinder head and the crankcase.

The injection pump with engine governor receives oil from the camshaft. The exhaust turbochargers of turbocharged engines are also connected to the oil system.

The oil can be extracted from engine and gearbox with the aid of a manually operated vane pump not mounted on the engine.

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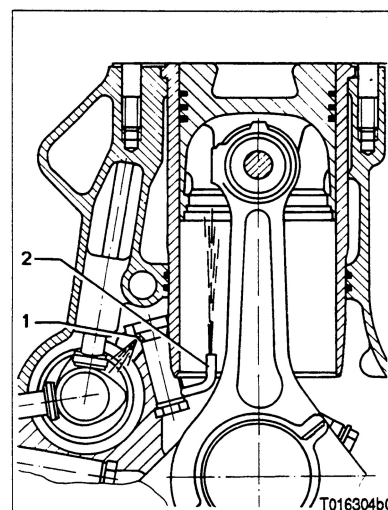
Piston Cooling and Cam Lubrication

An oil spray nozzle cools the piston crown. Oil is sprayed onto the piston before engine speed increases above idle. The banjo bolt and integral pressure relief valve that secures the piston cooling oil spray nozzle also supplies an oil mist to the cams of the camshaft.

Maintenance:

Oil change
Oil system filling
Oil level check
Oil sample extraction and analysis

} see page 67
}
} see Operation Manual M0 15110

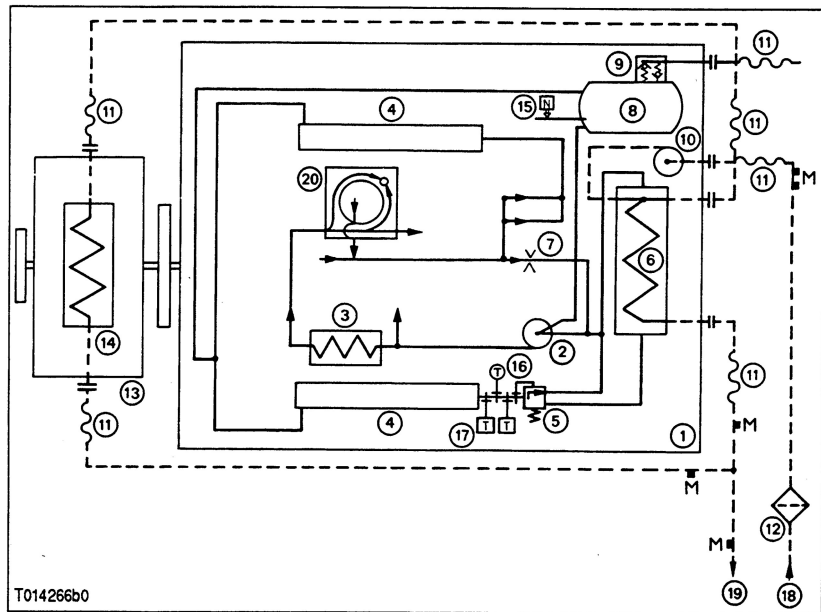


- 1 Spray nozzle for cam lubrication
- 2 Spray nozzle for piston cooling

Description of Engines

Coolant System with Plate-type Cooler V 183 AA 51/61/91

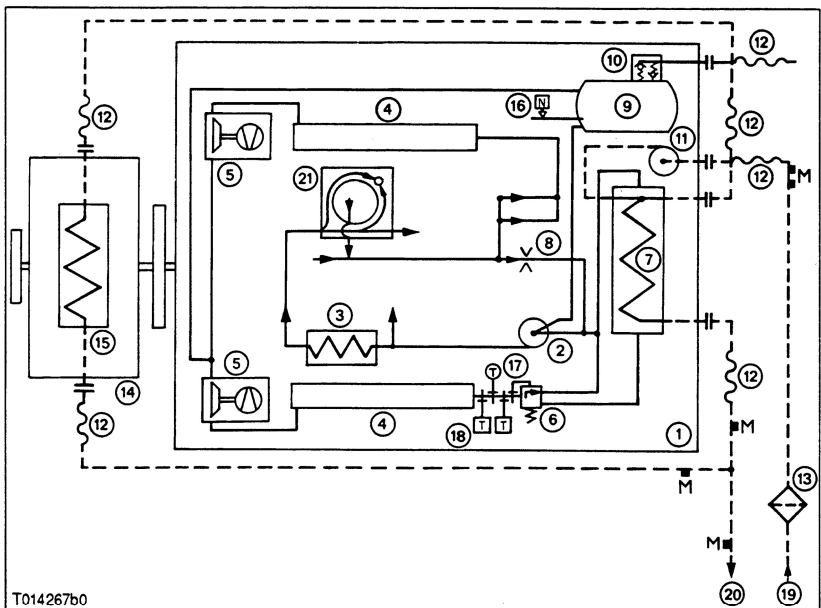
- 1 Engine
 - 2 Engine coolant pump
 - 3 Engine oil heat exchanger
 - 4 Exhaust manifold
 - 5 Coolant thermostat
 - 6 Coolant cooler (plate-type cooler)
 - 7 Restrictor
 - 8 Coolant expansion tank (coolant distributor housing)
 - 9 Breather valve
 - 10 Raw water pump
 - 11 Flexible connection
 - 12 Raw water filter
 - 13 Gearbox
 - 14 Gearbox oil heat exchanger
 - 15 Fluid level monitor
 - 16 Temperature transmitter
 - 17 Temperature monitor
 - 18 From raw water chest
 - 19 To outboard
 - 20 Cylinder
- M Measuring point
 — Engine coolant
 --- Raw water



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Coolant System with Plate-type Cooler 12V 183 TA 51/61/91

- 1 Engine
 - 2 Engine coolant pump
 - 3 Engine oil heat exchanger
 - 4 Exhaust manifold
 - 5 Exhaust turbocharger
 - 6 Coolant thermostat
 - 7 Coolant cooler (plate-type cooler)
 - 8 Restrictor
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